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    SCHOOL BUS ROUTE OPTIMIZATION
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SELECT CITY
1. Rajahmundry
2. Peddapuram
3. Samalkot

Select City: 1

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CITY: Rajahmundry
========================================
Direction: Home to School
Bus Capacity: 50
Total Students: 35
Capacity Check: Successful

--- NEAREST NEIGHBOR (GREEDY) ---
Route: Rajahmundry -> Gandhipuram 1st Street -> Danavaipeta -> Lalacheruvu ->
Pragati English Medium School, Kakinada
```

```
Total Distance: 72 km

--- BRANCH AND BOUND ---
Route: Rajahmundry -> Lalacheruvu -> Danavaipeta -> Gandhipuram 1st Street ->
Pragati English Medium School, Kakinada
Minimum Distance: 70 km

--- BRUTE FORCE VERIFICATION ---
Minimum Distance: 70 km

--- DISTANCE COMPARISON ---
Nearest Neighbor: 72 km
Branch and Bound: 70 km
Distance Saved: 2 km

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VERIFICATION
========================================
Branch and Bound result verified successfully!

One bus completes one trip.
No second bus or second trip is used.
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Do you want to check another city?
1. Yes
2. No
Enter choice:

=== Server Restarting, Please Run Again ===
```